Khairy Fathy Abd El-Sayed Sayed

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	Scopus	Google Scholar
Citations	**2062**	**3550**
h-index	**26**	**34**

https://www.scopus.com/authid/detail.url?authorId=24780869400
https://scholar.google.com/citations?user=gD0R0hQAAAAJ&hl=en
http://orcid.org/0000-0001-8871-5358
https://publons.com/researcher/1171925/khairy-sayed/metrics/
Researcher ID: E-6498-2015

EDUCATION:

Fulbright Research Scholar	Postdoctoral research scholar at City University New York 2024-2025 Advancing Sustainable Transportation: Exploring Cost-Effective Power Conversion Solutions for Electric Vehicles
Post Doctor	Control and Protection of Resilent Microgrid University of Ontario Institute of Technology Sept. 2016, Canada
PHD	PHD in Electrical Engineering, April 2013 Assiut University, Egypt
PHD Thesis	Analysis, design and control of stand-alone and grid connected PV systems
M.Sc.	M.Sc in Electrical Engineering, August 2007 Kyungnam University, South Korea
M.Sc. Thesis	A Study on Soft Switching PWM DC-DC Power Conditioning Converters for Energy Conversion Applications.
B.Sc.	B.Sc. in Electrical Engineering, Power and Machines Section, Assuit University, June 1997
B.Sc. Project	Voltage Flicker due to Arc Furnaces Load and Welding Machines Problems and Solutions

PROFILE

Name	:	Khairy Fathy Abd El-Sayed Sayed
Date of birth	:	24/6/1974
Nationality	:	Egyptian.
Marital Status	:	Married
Military Service	:	Finished
Home Address	:	Egypt,Assiut, 136 Beat Aila Awal. New Assiut City

Email Address : Khairy.f@gmail.com , khairy_sayed@eng.sohag.edu.eg
Website : https://staffsites.sohag-univ.edu.eg/khairy_abdelsaid

OBJECTIVES:

To work with a reputed institution as an Associate professor, that will provide me a good platform to utilize my teaching & administration skills and will help me to grow my career

RESEARCH INTERESTS

- Renewable Energy, Solar PV, Wind Energy , Fuel cell, Grid-connected power conditioner for renewable energy sources.
- Soft switching DC-DC power converter topologies, high frequency inverters and power factor correction, Grid-connected Inverters.
- Electric Vehicle and Smart grid
- Smart Energy Grids and Microgrids Planning, Protection, and Control
- Smart Green Buildings
- Competency based training and education

FUNDED RESEARCH PROJECTS

I participated in the following Research Projects

- Design of power conditioner systems for Fuel Cell implemented in Kyungnam University, 2006.
- Power conditioner for Solar Photovoltaic interface (Daihen Corporation, Japan), 2006.
- Soft Switching DC-DC power converter for Arc welding (Daihen Corporation, Japan), 2006
- Fast battery charger implemented in Korea Electro-technology Research Institute (KERI), 2007.
- High voltage magnetron power supply implemented in Korea Electro-technology Research Institute (KERI), 2007.
- A Stand-alone residential PV system (STDF Egypt, Project ID: ID1481, Oct. 1st, 2010 to Sep. 30th, 2011).
- A hybrid Solar-Wind Generation Based Micro-Grid for the Irrigation System of A Major Land Reclamation Project in Egypt-Case Study of the Toshka Project (STDF Egypt, USAID, Project Start Date: October 1st, 2010 – End Date: November 30th, 2013).
- Power Management Strategy for Battery Electric Vehicles used in Sohag University, 2020
- Microgrid technology applications for Majmaah University Campus. Majmaah University, the Deputyship for Research and Innovation, Ministry of Education in Saudi Arabia, for funding this research work through project number (IFP-2020-03), 2020.
- Implementation of Brushless Direct Current (BLDC) Motor Drive for Electric Vehicles, STDF from 18/8/2021 to 17/8/2024.

EXPERIENCE

Current job

July 2018 Associate professor, Department of Electrical Engineering, Sohag University, Egypt

Main Duties – Associate Professor of Electrical Engineering

- **Teaching and course development:** Deliver undergraduate and postgraduate courses in electrical engineering, and develop curricula, lectures, laboratory activities, and assessment methods.
- **Research and innovation:** Conduct and supervise research in electrical power systems, renewable energy, smart grids, power electronics, AI, and related areas.
- **Student supervision:** Supervise senior graduation projects, MSc and PhD students, providing technical guidance and research mentoring.
- **Academic leadership:** Participate in departmental committees, curriculum development, accreditation activities, and academic quality assurance.
- **Research dissemination:** Publish research in international peer-reviewed journals and conferences and contribute to research proposals and collaborative projects.
- **Laboratory and technical management:** Support the development and effective operation of electrical engineering laboratories, equipment, and simulation facilities.
- **Industry and professional engagement:** Establish collaborations with industry, research institutions, and professional organizations to support applied research and student training.
- **Academic mentoring:** Mentor junior faculty members and contribute to faculty development and continuous improvement of teaching and research activities.

May 2013 Assistant Professor, Department of Electrical Engineering, Sohag University, Egypt

Previous Job (industrial experience)

Electrical Engineer in Middle Egypt Electricity Distribution Company (August 1999 to September 2008)

- Working in Assiut Control center, DMS (SCADA) Project, Distribution Management System (DMS) operation Engineer, Protection, Maintenance, DTS dispatcher training simulator and switching management systems.
- Installation of Digital relays, Digital Meters, Remote Terminal Unit RTU
- Installation of SCADA system project (Installation works in Substations, Distribution points and Distribution Transformers, control room and UPS systems)

Duties as Control Engineer

Middle Egypt Electricity Distribution Company
(1999–2008)

⚡ SCADA System Operations

- Operated and monitored **Supervisory Control and Data Acquisition (SCADA)** systems to manage distribution grid performance in real-time.
- Diagnosed faults, responded to grid disturbances, and coordinated with field teams for fault isolation and restoration.

🔌 Electrical Network Monitoring & Control

- Oversaw **medium- and low-voltage distribution systems**, ensuring stable voltage levels and efficient power delivery across substations and feeders.
- Implemented control strategies to minimize technical losses and optimize load distribution.

🛠 Preventive Maintenance & Troubleshooting

- Conducted preventive maintenance routines on **control panels, circuit breakers, relays, and transformers**.

- Investigated and resolved system faults, tripping issues, and abnormal equipment behavior.

System Analysis & Reporting

- Collected and analyzed **grid performance data**, including load profiles, fault logs, and energy consumption patterns.
- Prepared technical reports for operational planning, reliability improvement, and performance benchmarking.

Control Panel Design & Upgrades

- Participated in **design and installation** of new control systems and retrofitting of outdated panels.
- Assisted in upgrading distribution panels to comply with safety and operational standards.

Team Coordination & Training

- Coordinated with dispatchers, engineers, and maintenance crews to ensure safe and efficient grid operations.
- Delivered **technical training sessions** to newly hired engineers on SCADA functions and protection systems.

Grid Optimization Projects

- Contributed to automation and **loss reduction projects**, such as capacitor bank control, feeder reconfiguration, and real-time demand monitoring.
- Assisted in deploying **smart metering pilots** and integrating remote control capabilities.

Vocational Education Development Expert

- In depth knowledge of the German dual vocational training system from more than 5 years ongoing cooperation with IP Consult and the State Academy Baden Württemberg for EDF's Integrated Technological Education Cluster (ITEC) project in Assiut
- 5 Year experience in teacher Coaching and teacher training for Vocational education
- Experience in applying and adapting the German dual vocational training system to the Egyptian context
- Publication in competency based education journal:

  Development of competency-based training system in Assiut-ITEC: A case study
  An Extensive Model for Implementing Competency-Based Training (CBT) in TVET Teacher Training System for Assiut-ITEC, Egypt.

Associate professor in Electrical power Engineering

- Extensive teaching and research experience in **power systems, electrical machines, power electronics, dynamic systems, and advanced control**, with applications in renewable energy, electric transportation, and intelligent energy systems.
- Research and development of **advanced power electronic converters and control systems** for renewable-energy applications, electric vehicles, battery energy-storage systems, and electrified transportation.
- Research and development in **aerospace and electric-vehicle dynamics and control**, including dynamic modeling, advanced control strategies, model predictive control (MPC), intelligent control, electric propulsion, hybrid-electric propulsion, and energy-management systems.
- Experience in the **modeling, analysis, design, and control of electric machines and drives**, including BLDC and PMSM drives and their applications in electric vehicles and More Electric Aircraft.
- Experience in the **design and analysis of electrical substations, medium- and low-voltage (MV/LV) distribution networks, electrical installations, and building energy systems**, with emphasis on reliability, efficiency, power quality, and safe operation.

- Experience in **energy infrastructure planning and design**, including the development of reliable, efficient, and sustainable energy systems while considering **technical performance, economic feasibility, environmental impact, energy efficiency, and system reliability**.
- Experience in **embedded control, MATLAB/Simulink modeling, real-time simulation, and hardware-oriented implementation** of advanced control algorithms for power electronic, electric-drive, energy, and transportation systems.
- Interdisciplinary research experience integrating **artificial intelligence, machine learning, optimization, control systems, power electronics, energy storage, and electric propulsion** for intelligent and autonomous engineering systems.

COURSES TAUGHT

- Electrical Machines
- Power Electronics
- Power system analysis
- Power system protection
- Electrical Circuits
- Electrical Measurements
- Electromagntic fields
- High Voltage Engineering
- Energy conversion and utilization
- Energy Economics
- Electrical Installation & Drawing.
- Smart Energy Grids
- Microgrids Planning, Protection, and Control
- Renewable energy
- Motor Drives
- Energy coversion and utilization

AWARDS AND HONORS

1- Best Student Paper Award at the 57 annual conference of the International Appliances Technical Conference IATC2006 held at the Crowne Plaza Chicago, USA, March 27-29, 2006 for the paper entiteld " Dual Mode Pulse Modulated Soft-Switching series Load Resonant Inverter with active Edge Resonant Snubber for High Frequency Induction Heating Appliances .
2- Sohag University Encouragement Awards in the field of engineering science 2016-2017.
3- Peer review awards certificate (Mendeley 2018)
4- Fulbright Research Scholar Award, 2024-2025

PATENT

Smart DC-DC Converter (SDDC) for DC Energy Harvesting, US patent, under review, submitted 18-APR-2016, Application Number 62323999.

INVITED TALK

Recent Trends in Power Electronics, Present Status and Future Prospects.
University of Ontario Institute of Technology, Faculty of Energy Systems and Nuclear Science, Friday April 8th,2016.
Organizing a workshop for the generation of hybrid energy for wind and solar for irrigation system in Toshka, Assiut University on 30/11/2013.
Essentials and Quality Criteria for Technical ocational Education in ITEC Assiut Workshop, 12. / 13. November 2018 Assiut

Artificial Intelligence for Electric Vehicles: Intelligent Battery Management and Power Electronics Technologies, **Xi'an Jiaotong University Summer School, July 2026.**

REVIEWER FOR INTERNATIONAL JOURNALS

1) Reviewed papers submitted to the Journal of IET Electric Power Applications, the Journal of IET Renewable Power Generation, the IEEE Transactions on Industry Applications, the IEEE Transactions on Power Electronics, IEEE Transactions on Industrial Electronics the Journal of Electric Power Components and Systems, Journal of Power Electronics, International Transactions on Electrical Energy Systems.
2) Reviewed several papers submitted to IEEE International Conferences.

COMMUNICATION AND PRESENTATIONS SKILLS

International conferences Speaking

Presentation of my papers in international conferences and in Domestic conferences (in Korea)
I have presented my papers in the following international conferences

1- PEDS 2005 (International Conference on Power Electronics and Drive Systems), Kuala Lumpur, Malaysia.
2- ICIT 2005 (International Conference on Industrial Technology), Hong Kong.
3- PEMD 2006 (Power Electronics and Machine Drive Conference), Dublin, Ireland.
4- IPEMC 2006 (International Power Electronics and Motion Control Conference), Shanghai, China.
5- ICEMS 2006 (International Conference of Electrical Machines and Systems), Nagasaki, Japan.
6- PESC 2006 (Power Electronics Specialists Conference), Korea.
7- ICEE 2006 (International conference of Electrical Engineering), Korea.
8- Power Electronics Technology Oct. 24-26, 2006, Long Beach, California, USA.
9- IEE Japan Annual Conference 2007, Toyama, Japan.
10- IEEE Middle East Power System Conference 2008, Aswan, Egypt.
11- IEEE ICEMS 2009, Tokyo, Japan
12- IEEE ICPE 2011, Korea.
13- MEPCON 2012, Egypt
14- INTELEC 2013, Germany
15- IPEC 2014 Hiroshima, Japan
16- SEGE 2016 Oshawa, Canada

COMPUTER SKILLS:

- Programming Language: Fortran, Visual Basic, C++ Programming languages

- Matlap, Simulink, PSIM6.1, MathCAD, Pspice, Orcad, Simplorer, Maxwell 2D
- AutoCAD, Designer
- Operating Systems: Unix operating system (User and Administration) nd Windows NT.
- SCADA/DMS software
- Microsoft office(Word, Excel, Power Point, Access) and Microsoft project.
- PLC programming, PIC and DSP microcontroller programming
- Arduino, PIC microcontrollers and PLC

RELAVANT SKILLS TRAINING

- Quality Assurance (QA) Training Course – Egypt, 2018
- Competency based Engineering Education, and the State Academy Baden Württemberg, 2017
- ToT training Course, Sohag University, 2016

SUPERVISION OF GRADUATION PROJECTS

- Multi-functions Surveying RC Remote Controlled Unmanned Aerial Vehicle UAV 2013
- Smart Grid Communications using Wireless Communications, 2014.
- Vehicle Tracking Using GSM System 2014
- RFID Smartcard based Access Control System, 2015
- PV powered Irregation system using wirless commuincations 2016-2017
- Web-SCADA for Monitoring and Controlling Hybrid Wind-PV Power System 2017-2018
- Development of an Automated INDUSTRIAL Robot 2018-2019
- Design and Implementation of elevator control system 2018-2019
- Speed control of induction motor, 2020.
  - Electrical distribution system for specialized hospital, 2022.
  - Design of Grid-Connected Large-Scale Photovoltaic Power Plants, 2023
  - IoT Based Stepper Motor Position Control for Industrial Automation, 2024

PROFESSIONAL AFFILIATIONS

IEEE member
KIEE member (Korean Institute of Electrical Engineers)
KIPE member(Korean Institute of Power Electronics)
Egyptian Engineers Syndicate
Member, Egyptian Society of Engineers
Member, International Association of Engineers (IAENG)

PUBLISHED BOOKS

1. Ahmed G. Abo-Khalil, Ramesh C. Bansal, Khairy Sayed, Ahmed Almakky, Chapter 2 - Emerging electric vehicle technologies: Topologies, charging infrastructure, grid stability, and economic impacts, Editor(s): Abhishek Kumar, Ramesh C. Bansal, Praveen Kumar, Xiangning HE, Handbook on New

Paradigms in Smart Charging for E-Mobility, Elsevier, 2025, Pages 53-79, ISBN 9780323952019, https://doi.org/10.1016/B978-0-323-95201-9.00002-0.

2. Smart Energy Grid Book – Chapter 18, " SCADA and Smart Energy Grid Control Automation", Pages 481-514, Elsevier 2016.
3. Building Energy Management Systems (BEMS), (Pages: 15-81), June 2018, In book: Energy Conservation in Residential, Commercial, and Industrial Facilities, Published 2018 by John Wiley & Sons, DOI: 10.1002/9781119422099.ch2
4. Book Chapter, Microinverter Systems For Energy Conservation In Infrastructures, in book: Energy Conservation in Residential, Commercial, and Industrial Facilities, (Pages: 125-202), June 2018 , Published 2018 by John Wiley & Sons, DOI: 10.1002/9781119422099.ch5
5. Book Chapter, Wind Power Plants Control Systems based on SCADA system , in book: Control and Operation of Grid-Connected Wind Energy Systems (Green Energy and Technology) 1st ed. 2021 Edition, Publisher: Springer International Publishing, by Ali M. Eltamaly (Editor), Almoataz Y. Abdelaziz (Editor), Ahmed G. Abo-khalil (Editor), eBook ISBN: 978-3-030-64336-2, March 2021, DOI: 10.1007/978-3-030-64336-2_2
6. Different Approaches for Efficiency Optimization of DFIG Wind Power Generation Systems, Khairy Sayed, Ahmed G. Abo-Khalil, and Ali M. Eltamaly, March 2021, DOI: 10.1007/978-3-030-64336-2_2. In book: Control and Operation of Grid-Connected Wind Energy Systems Publisher: Springer
7. Design and Optimization of a Hybrid Energy Storage System for Standalone Renewable Energy Systems, By Ahmed G. Abo-Khalil, Ramesh C. Bansal, Khairy Sayed, BookMicrogrid Handbook, 1st Edition, First Published 2025, Imprint CRC Press. Pages 24
8. Hybrid and Advanced Energy Storage Systems: Integration, Applications, and Future Trends, Authors, Khairy Sayed, Mahmoud Aref, Ahmed G Abokhalil, Mahmoud A Mossa, Publication date 2025/4/17. Book: Energy Storage Devices-A Comprehensive Overview, Publisher: IntechOpen
9. Khairy Sayed, Transformers and Magnetic Circuits (Arabic), pp. —, 2011. ISBN: 977-440-116-8, Deposit No. 2011/19720
10. Khairy Sayed, Basics of Power Systems (Arabic), pp. —, 2010. ISBN: 977-440-105-3, Deposit No. 2203/2010
11. Khairy Sayed, Maintenance and Testing of Power Transformers (Arabic), pp. —, Al-Dar Al-Alamiya Publishing, 2018. ISBN: 978-977-440-315-5, Deposit No. 27534/2017
12. Khairy Sayed, Measurement and Protection in Power Systems (Arabic), pp. —, Al-Dar Al-Alamiya Publishing, 2019. ISBN: 978-977-440-317-9, Deposit No. 27536/2017
13. Khairy Sayed, Design and Analysis of Solar Photovoltaic Systems (Arabic), pp. —, Al-Dar Al-Alamiya Publishing, 2019. ISBN: 978-977-440-316-2, Deposit No. 27535/2017
14. Khairy Sayed, Power Transformers Engineering (Arabic), pp. —, Al-Dar Al-Alamiya Publishing, 2018. ISBN: 978-977-440-313-1, Deposit No. 26823/2017
15. Khairy Sayed, Electrical Energy Generation (Arabic), pp. —, Al-Dar Al-Alamiya Publishing, 2019. ISBN: 978-977-440-314-8, Deposit No. 27533/2017

SUPERVISION OF MSC. STUDENTS

MSc Students – finished:

1. Analysis of a Multifunction DSP-based-Numerical Relay for Distribution Systems with Distributed Generators, Mohsen Badr, Faculty of Engineering, Assiut University, 2015.
2. Control Techniques For DC-DC Converters With Improved Performance, Hedra Mahfouz, Faculty of Industrial Education, Sohag University, 2015.

3. Harmonic Measurements And Analysis of Welding Machines, Wesam Arfa, Faculty of Industrial Education, Suez Canal University, 2015.
4. Optimal Energy Storage system for Microgrid in the Presence of Renewables, Amal Mohamed Abd Elhamed, Faculty of Industrial Education, Sohag University, 2016.
5. Design and Implementation of Fly-back Converter for Batteries Charging Applications, Rasha Sayed, Faculty of Industrial Education, Sohag University, 2015.
6. Load Frequency Control and Power System Stabilizer, Hosam Hassan Ali, Faculty of Industrial Education, Sohag University, 2017.
7. Performance Improvement of Stand-alone Hybrid PV / Storage power System, Alaa Abd-Elsamee, Faculty of Industrial Education, Sohag University, 2017.
8. Design and Performance Analysis of Series Resonant Inverters, Heba Elzohry, Faculty of Industrial Education, Sohag University, 2019.
9. Harmonics control and performance improvement of a grid connected photovoltaic system, Ikhlas Mohamed, Faculty of Industrial Education, Sohag University, 2019.
10. Energy Management and Control of Flywheel Energy Storage Systems in Microgrid, Ismail Hassan, Faculty of Engineering, Aswan University, 2020.
11. Intelligent Fault Location Techniques in Electrical Power Networks, Mohamed Bahy, Faculty of Engineering, Assiut University, 2020.
12. Protection of distribution transformer in presence of Renewable Energy Sources, Amal Edwar, Faculty of Industrial Education, Sohag University, 2021.
13. Control for grid connected Hybrid photovoltaic and Wind Energy generation system , Mohamed Mourad, Faculty of Engineering, Assiut University, 2022.
14. Hesham Mohamed Abd El-monam, "Analysis and Control of DFIG Driven by a Variable Speed Wind Turbine", 2022.
15. Mohammed Ail Abd-el Zaher Hassan, "Design and implementation of High-efficiency Resonant Converter for led Drivers", 2022.
16. Rabaa Mahmoud, Improvement of solar cell efficiency depending on control the parameters of fabrication stages, Faculty of Engineering, Assiut University, 2022
17. Hanan Hamdy, Design and Analysis of High Frequency Inverter for metal forming applications, Assiut University, 2023.
18. Advanced Control Techniques for Improvement of a Heliostat Sun Tracking System Performance, Khaled, 2024.
19. An efficient Methods for Deeplearning Based Automatic Modulation Classification, Mohamed Elsagheer, 2024
20. Abdelhady Ramadan, Performance Enhancement of Permanent magnet Synchronous Generators for Wind Turbines Using Artificial Intelligence, Faculty of Engineering, Sohag University, 2024.
21. Nagwa Albadry, Application of Modern Optimization Algorithms for Performance Enhancement of the Doubly Fed Induction Generators Based on Wind Turbines, 2025
22. Sayed Taha, Design, Operation and Control of a DC Microgrid based on Renewable Energy Sources, 2026

PHD Students (Finished)

1- Hedra Mahfouz, Analysis, Design and Control in Electric Vehicles, June 2020.
2- Rasha Kassem, Performance Evaluation of Induction and Permanent-Magnet Synchronous Machines for Loss Minimization during Drive Cycles Using Fuzzy Logic Controllers in Electric Vehicle, July 2020.
3- Heba Elzohry, Implementation of Brushless Direct Current (BLDC) Motor Drive for Electric Vehicles. Sohag University, 2024.

4- Mohamed AbdelRahman. Control Techniques for Optimal Energy Management in Microgrid ,Sohag University , 2024.

5- Islam Kamal, Design and Analysis of Multi-Output Power Supply for Metal Forming Applications,2026.

Ongoing Msc.

1. Monitoring and Rationalizing Energy Consumption of Home Electric Appliances, Allam Aldeep, Faculty of Industrial Education, Sohag University.
2. Ahmed Mostafa, Implementation Of DC Microgrid With Efficient Energy Management Of Multiple Energy Sources
3. Enhancing Stability in Low-Inertia Power Grids Based on Advanced Controllers Integrated with Energy Storage Technologies
4. Advanced Design and Energy Management of Smart Grids for Green Fuel Production
5. *Mohamed Mourad, Digital Twin for Real-Time Energy Market Simulation: Modeling Market Behavior, Strategies, and Peer-to-Peer Energy Trading*

RESEARCH ACHIEVEMENT LIST

Publications List Dr Khairy Sayed

Journal papers

1. El Hamid, A.M.A., Elzohri, H.H. & Sayed, K. Integrated machine learning forecasting and grey wolf optimization for optimal operation of virtual power plants in smart distribution networks. *Sci Rep* **16**, 24535 (2026). https://doi.org/10.1038/s41598-026-65123-1
2. Khairy Sayed, Ahmed G. Abokhalil, Mahmoud Aref, Mishari Metab Almalki, Mahmoud A. Mossa, Digital twin–enabled urban systems for sustainable and resilient futures: A global urban science perspective, Sustainable Futures, Volume 11, 2026, 101902, ISSN 2666-1888, https://doi.org/10.1016/j.sftr.2026.101902.
3. Mosbah, M.A., Abokhalil, A. & Sayed, K. Toward adaptive control power sharing and bus voltage regulation for DC microgrids. *Sci Rep* **16**, 13395 (2026). https://doi.org/10.1038/s41598-026-47219-w
4. Ahmed G. Abo-Khalil, AlAmir Hassan, Khairy Sayed, HVDC Systems and Renewable Energy in a Comparative Study of Technologies and Applications, e-Prime – Nexus of Electrical, Electronic, and Intelligent Engineering, 2026, 201167, ISSN 3117-5112, https://doi.org/10.1016/j.eprime.2026.201167.
5. Khairy Sayed, Hebatallah H. ElZohri, Ahmed G. Abo-Khalil, Mahmoud Aref, A digital twin–enabled measurement and instrumentation framework for ultra-accurate short-term voltage prediction in lithium-ion batteries for electric vehicle applications, Next Energy, Volume 12, 2026, 100612, ISSN 2949-821X, https://doi.org/10.1016/j.nxener.2026.100612.
6. Alobaid, M., Abo-Khalil, A.G. & Sayed, K. AI-enhanced techno-economic and environmental optimization for nearly zero-energy building retrofitting: a case study of university campus. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-41747-1
7. Abo-Khalil, A.G., Sayed, K., Mbungu, N.T. *et al.* Reinforcement learning driven adaptive active frequency drift for fast and reliable islanding detection. *Sci Rep* **16**, 876 (2026). https://doi.org/10.1038/s41598-025-31417-z
8. Hebatallah H. Elzohri, Khairy Sayed, Mohamed Mourad, Ahmed G. Abo-Khalil, Advancing solar PV power plants: The role of artificial intelligence in optimization, sustainability, and cost efficiency, Measurement: Digitalization, Volume 6, 2026, 100027, ISSN 3050-6441, https://doi.org/10.1016/j.meadig.2026.100027.
9. Nermin M. Salem, Khairy Sayed, Mohammed M. Elsayed, David Kirakosian, Ahmed A. Mohamed, Short-term discharge-based state of health estimation of lithium-ion batteries, Energy Reports, Volume 15, 2026, 108931, ISSN 2352-4847, https://doi.org/10.1016/j.egyr.2025.108931.
10. Khairy Sayed, Mohammed M. Elsayed, Ahmed Mohamed, Ahmad Eid, Key performance indicators for resiliency assessment in power systems with renewable energy and electric vehicles integration, Renewable and Sustainable Energy Reviews, Volume 225, 2026, 116135, ISSN 1364-0321, https://doi.org/10.1016/j.rser.2025.116135.

11. Mustafa Abu-Zaher, Ke Shen, Mokhtar Aly, Mohamad Abou Houran, Khairy Sayed, Ahmed G. Abo-Khalil, Alaaeldien Hassan, Power quality enhancement in grid-tied photovoltaic systems through artificial neural network-based current control strategies, International Journal of Electrical Power & Energy Systems, Volume 172, 2025, 111362, ISSN 0142-0615, https://doi.org/10.1016/j.ijepes.2025.111362.
12. Hassan, I., Sayed, K., Alanazi, M.D. *et al.* A comprehensive review of power converter topologies and control techniques for welding power supply applications. *Discov Appl Sci* **7**, 1036 (2025). https://doi.org/10.1007/s42452-025-07180-1
13. Alaa A. Mahmoud, Khairy Sayed, Amil Daraz, Yogendra Arya, Mohamed Khamies, PO-optimized cascaded FOIAN-PTD strategy for frequency control of wind-PV-thermal power system with energy storage systems, Engineering Science and Technology, an International Journal, Volume 70, 2025, 102173, ISSN 2215-0986, https://doi.org/10.1016/j.jestch.2025.102173.
14. Mahmoud Aref, Mahmoud A. Mossa, Emad Abdelkarim, Khairy Sayed, Mishari Metab Almalki, Alaa F.M. Ali, Enhancement of the operating time of the overcurrent relay of the distribution network with high-level penetration of renewable energy sources, Results in Engineering, Volume 26, 2025, 104859, ISSN 2590-1230, https://doi.org/10.1016/j.rineng.2025.104859.
15. K. Sayed, M. M. Elsayed and A. Mohamed, "Feasibility and Economic Assessment of a PV-Wind Hybrid Charging Station in the Bronx, NY," in *IEEE Access*, vol. 13, pp. 61841-61861, 2025, doi: 10.1109/ACCESS.2025.3555158.
16. Gronfula, Mohammed, and Khairy Sayed. 2025. "AI-Driven Predictive Control for Dynamic Energy Optimization in Flying Cars" *Energies* 18, no. 7: 1781. https://doi.org/10.3390/en18071781
17. Khairy Sayed, Mahmoud Aref, Mishari Metab Almalki, Mahmoud A. Mossa, Optimizing fast charging protocols for lithium-ion batteries using reinforcement learning: Balancing speed, efficiency, and longevity, Results in Engineering, Volume 25, 2025, 104302, ISSN 2590-1230, https://doi.org/10.1016/j.rineng.2025.104302.
18. Mohamed Khamies, Khairy Sayed, Omar Alrumayh, Abdulaziz Almutairi, Alaa A. Mahmoud, Advanced active disturbance rejection control for enhancing frequency stability in low-inertia power grids linked with virtual inertia applications, Heliyon, Volume 11, Issue 4, 2025, e42556, ISSN 2405-8440, https://doi.org/10.1016/j.heliyon.2025.e42556.
19. K. Sayed *et al*., "Feasibility Study and Economic Analysis of PV/Wind-Powered Hydrogen Production Plant," in *IEEE Access*, vol. 12, pp. 76304-76318, 2024, doi: 10.1109/ACCESS.2024.3406895.
20. Mesbah, M.A., Sayed, K., Ahmed, A., lmalki, M.M., Alghamdi, T.A.H., "A Distributed Architecture of Parallel Buck-Boost Converters and Cascaded Control of DC Microgrids-Real Time Implementation", IEEE Access, 2024, 12, pp. 47483–47493
21. Sayed, K.; El-Zohri, H.H.; Ahmed, A.; Khamies, M. Application of Tilt Integral Derivative for Efficient Speed Control and Operation of BLDC Motor Drive for Electric Vehicles. *Fractal Fract.* **2024**, *8*, 61. https://doi.org/10.3390/fractalfract8010061
22. Mesbah, M.A.; Sayed, K.; Ahmed, A.; Aref, M.; Elbarbary, Z.M.S.; Almuflih, A.S.; Mossa, M.A. Adaptive Control Approach for Accurate Current Sharing and Voltage Regulation in DC Microgrid Applications. *Energies* **2024**, *17*, 284. https://doi.org/10.3390/en17020284
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